Dilan Karaguler

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Summary

Mathematics Ph.D. candidate specializing in graph machine learning and topological data analysis, with two Intel internships building graph-based ML pipelines for semiconductor layout analysis and device classification. Experienced in turning new mathematical methods into working Python and PyTorch tools.

Technical Skills

Programming: Python, R, SQL
Methods: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation
Libraries: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM
Tools: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, LaTeX
Domain Knowledge: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

Education

Michigan State University — East Lansing, MI, US
Ph.D. in Mathematics (Candidate) — August 2022 – May 2027 (Expected)
- Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

Middle East Technical University (METU) — Ankara, Türkiye
M.S. in Mathematics — September 2019 – September 2021
- Thesis: *A Survey on Multidimensional Persistence Theory*
- Advisor: Assoc. Prof. Semra Pamuk

Middle East Technical University (METU) — Ankara, Türkiye
B.S. in Mathematics, Honor Student — June 2014 – June 2019

Industry Experience

Software Engineer Intern — May 2026 – August 2026
Intel Corporation — Oregon, US
- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

Graduate Technical Intern — May 2025 – August 2025
Intel Corporation — Oregon, US
- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

Research Experience

Graduate Researcher August 2022 – Present
Michigan State University East Lansing, MI, US

- Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).
- Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

Research Mentor May 2023 – June 2023
Topology RTG, Michigan State University East Lansing, MI, US

- Mentored an undergraduate researcher on a project in Morse theory: project design, methodology and research presentation; the student went on to major in mathematics.

Projects

Persistent Mayer Homology with Dynamic Mode Decomposition April 2026
Python, NumPy, SciPy, GUDHI GitHub

- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

Homology-Fed Neural Network Framework November 2024
Python, PyTorch, GUDHI, NumPy GitHub

- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

Cellular Automata on Curved Surfaces March 2024
Python, NumPy, Matplotlib GitHub

- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

Publications

D. Karaguler and G.-W. Wei. *Mayer Path Homology.* Foundations of Data Science, accepted (2026). doi:10.3934/fods.2026025, arXiv:2605.16525.

D. Karaguler. *A Survey on Multidimensional Persistence Theory.* M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians.* In preparation (2026).

Leadership & Awards

Excellence in Teaching Award, Department of Mathematics, MSU May 2025
Graduate Representative, NatSci Student Advisory Council, Michigan State University 2025 – 2026
Graduate Student Representative, Graduate Studies Committee, Department of Mathematics, MSU 2024 – 2025
President / Board Member, Association for Women in Mathematics, MSU Chapter August 2022 – August 2025
Lead Teaching Assistant, Department of Mathematics, MSU (mentoring and observing TAs) August 2024 – Present
Organizing Committee Member, Graduate Student Topology and Geometry Conference April 2024